

Total No. of Questions : 8]

SEAT No. :

PE-5955

[Total No. of Pages : 2

[6583]-333

**T.E. (Computer Engineering)
(Internet of Things)(Honors)**

**EMBEDDED SYSTEMS AND INTERNET OF THINGS
(2019 Pattern) (Semester - V) (310601)**

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Figures to the right side indicate full marks.
- 3) Neat diagrams must be drawn wherever necessary.
- 4) Assume suitable data, if necessary.

- Q1)** a) Briefly explain the principles of sensors. [6]
b) Define Actuators? Mention the different types of Actuation systems. [6]
c) Write the difference between Arduino and Raspberry Pi and Beagle Bone? [8]

OR

- Q2)** a) Define sensor. Classify different types of sensors. [6]
b) Explain about the different mechanisms used in rotary actuators. [6]
c) Write a note on following sensors. [8]
i) Active and passive
ii) Analog and digital
iii) Contact and no-contact
iv) Absolute and relative

- Q3)** a) Why is testing and validation particularly important in embedded systems, and what tools assist with it? [4]
b) Draw the architecture of SDLC and explain. [6]
c) What are common challenges in embedded application development and how to overcome them? [8]

OR

- Q4)** a) Explain SDLC components in detail. [4]
b) What are common security challenges in Embedded system? [6]
c) How can real-time constraints complicate embedded application development? [8]

P.T.O.

- Q5) a)** List the Main stages in the IoT architecture? And Explain. [6]
- b)** List and explain various application areas of IoT. [6]
- c)** Explain the role of sensors and actuators in an IoT system. [4]

OR

- Q6) a)** What is an IoT device? Give examples of different types of IoT devices. [6]
- b)** List and explain the enabling technologies for IoT [6]
- c)** How can interoperability issues create risks in IoT systems? [4]

- Q7) a)** How does NFC differ from RFID in terms of range and security features? [6]
- b)** What are the main IoT communication models? Explain each with an example. [6]
- c)** Define MQTT. What are its main features and typical use cases in IoT? [4]

OR

- Q8) a)** Describe the security features of MQTT, CoAP, XMPP and AMQP. [6]
- b)** Compare Bluetooth, BLE, Zigbee, Z-Wave, NFC, RFID and Wi-Fi in terms of range, data rate, power consumption and topology. [6]
- c)** What is XMPP? How is it used in IoT and what are its security mechanisms? [4]

